

Positive Rewards Alone Enable Bipedal Standing for Unitree Go2 in Isaac Lab

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Abstract—Reward design is a key challenge in reinforcement learning for robot control, especially for behaviors that require both large posture changes and stable balance. This paper studies RL-based bipedal standing for the Unitree Go2 in NVIDIA Isaac Lab. We compare a positive-rewards-only design with a mixed design that also includes penalty terms for unstable motion. Our results show that two positive reward terms are sufficient to learn the target behavior, while additional penalties provide little benefit and can hinder training. Instead, safety and feasibility are enforced through termination conditions. With this design and PPO in `skrl`, the learned policy maintains continuous bipedal standing for 9.72 out of a 10s evaluation window. These results suggest that, for this task, positive reward shaping combined with carefully designed termination rules is more effective than adding multiple penalty terms.

I. INTRODUCTION

Recent advances in artificial intelligence have accelerated progress in robotics, enabling increasingly capable systems for locomotion, manipulation, and whole-body control in both simulation and the real world [1], [2]. Among these methods, reinforcement learning (RL) is particularly attractive for robot control because it can learn adaptive policies directly from interaction, making it well suited to high-dimensional and contact-rich tasks such as legged locomotion [1]. However, one of the main practical difficulties in applying RL to robotics is reward function design [3].

In practice, the reward function is typically written as a weighted sum of *reward terms*, where each term measures one aspect of behavior, such as posture, height, smoothness, or safety. Because RL is reward-driven, even small changes in reward design can substantially change the learned policy.

In this work, we study reward design for a challenging legged robotics task: training a Unitree Go2 quadruped in NVIDIA Isaac Lab to rise from a quadrupedal stance to stable bipedal standing on its hind legs. Isaac Lab is particularly suitable for this study because its manager-based framework modularizes the main environment components, allowing us to vary the reward and termination design while keeping the remaining task setup unchanged [4].

Our central question is whether stable bipedal standing requires both positive rewards and penalty terms, or whether

a smaller set of positive rewards is sufficient when combined with appropriate termination rules. We find that positive rewards alone can learn the target behavior, whereas additional penalty terms provide little gain and may even degrade performance. The contribution of this paper is therefore a compact experimental study showing that, for this task, positive reward shaping plays a more effective role than adding multiple penalty terms.

II. METHODOLOGY

We formulate bipedal standing for the Unitree Go2 as a reinforcement learning control task in Isaac Lab. At each step t , the robot observes its state s_t , applies an action a_t , and receives a scalar reward r_t . The objective is to learn a policy $\pi(a_t | s_t)$ that maximizes the expected return

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right] \quad (1)$$

Here, the reward serves as feedback on how desirable the current behavior is with respect to the task. In our setting, the observation space contains proprioceptive information, including base linear and angular velocity, projected gravity, joint positions, joint velocities, and previous actions. The action space consists of 12 continuous commands, one for each actuated joint.

The reward function is defined as

$$R_t = \sum_i w_i r_i(s_t, a_t) \quad (2)$$

where each reward term measures one aspect of the desired behavior. We compare two reward designs. The first uses only positive rewards that encourage the target standing behavior, specifically height tracking and lift-up reward. The second augments this design with penalty terms for abrupt action changes, large joint velocities, and large joint accelerations and etc. The reward terms and weights are summarized in Table I. Our reward design is guided by prior work on bipedal skill learning for quadrupedal robots. Specifically, we draw on a study [5] implemented in the earlier Isaac Gym framework and validated on a Xiaomi robot dog. Although the simulator framework and robot platform differ from ours, we use the reward-term structure and its weight assignment as a design

<https://github.com/Wenlu057/robot-dog-standing>.

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reference, and adapt them to the Isaac Lab manager-based setting and the Unitree Go2.

In addition to rewards, we use termination conditions to enforce safety and feasibility constraints. An episode is terminated when the robot violates a joint-limit condition, exhibits unstable hind-foot motion, or produces an excessively abrupt joint change. These conditions are active only during the early phase of an episode, where unstable behaviors are most likely to appear. The termination terms are listed in Table II.

For policy learning, we use the `skrl` library and train with Proximal Policy Optimization (PPO), a standard algorithm for continuous-control RL. To keep the study focused on reward design, we use the default PPO hyperparameters in all experiments.

III. EXPERIMENTAL RESULTS

Our experiments indicate that two positive reward terms are sufficient to learn the target behavior. In contrast, adding penalty-based reward terms yields little improvement and can hinder learning. We therefore treat these constraints as termination conditions: if the robot violates a safety or feasibility constraint, the episode is terminated and reset. The reward terms and weights are listed in Table I, and the termination conditions are summarized in Table II.

We further find that termination timing is important. All three termination conditions are enforced only during the first 1000 simulation steps of each episode.

For training, we use the `skrl` library with PPO, a standard baseline for continuous-control RL. Default PPO hyperparameters are used in all experiments; hyperparameter tuning is left for future work.

With positive rewards and termination conditions alone, the learned policy achieves stable bipedal standing. Figure 1 shows that the robot maintains success for 9.72 out of 10 seconds on average. A representative snapshot in Isaac Sim is shown in Fig. 2.

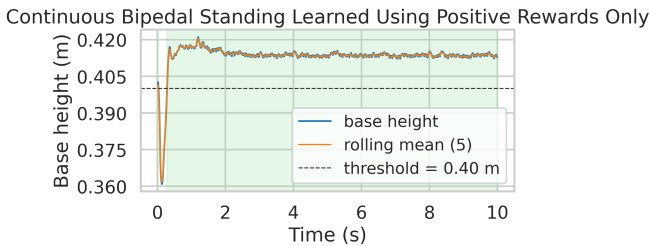


Fig. 1. Base height during evaluation with positive rewards only. The learned policy maintains stable bipedal standing above the 0.40 m threshold for nearly the full 10 s horizon.

IV. CONCLUSION

We hope the practical design choices and findings reported here can serve as a useful reference for future robotics control research using the newer Isaac Lab environment.

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TABLE I
REWARD TERMS AND WEIGHTS.

Term	Equation	Weight
Upright height	$1 - z_t - z^* $	1
Lift-up height	$\text{clip}\left(\frac{z_t - 0.15}{0.52 - 0.15}, 0, 1\right)$	0.8
Action smooth.	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.03
Joint vel.	$\ \dot{\mathbf{q}}\ _2^2$	-10^{-4}
Joint accel.	$\ \ddot{\mathbf{q}}\ _2^2$	-2.5×10^{-7}
Joint limit	$\sum_j \phi(q_j)$	-10
Foot clr.	$\sum_{f \in \mathcal{H}} (h_f^* - h_f)^2 (1 - c_f)$	-300
Foot slip	$\sum_f \mathbb{K}[z_f < \epsilon] (\ v_f\ ^2 + \omega_f^2)$	-50
Undesired cont.	$\sum_b \mathbb{K}[F_b > \tau]$	-2.0
Q diff	$\mathbb{K}[t \leq 30] \ q_t^{\text{ref}} - q_t\ _2^2$	-1.0
Torque limit	$\text{clip}\left(\sum_j \tau_j^{\text{app}} - \tau_j^{\text{cmd}} , 0, 10\right)$	w_7

Notation: z_t is the current base height; z^* is the target base height; $z_{\min}$ and $z_{\max}$ are the lower and upper heights used for normalization; $\mathbf{a}_t$ and $\mathbf{a}_{t-1}$ are the current and previous action vectors; $\dot{\mathbf{q}}$ and $\ddot{\mathbf{q}}$ denote joint velocities and accelerations; $\phi(q_j)$ denotes the penalty applied when joint j approaches its position limits; h_f is the current height of hind foot f ; h_f^* is its target clearance height; c_f is the desired contact state; and $\mathcal{H}$ denotes the hind feet; v_f denotes horizontal foot velocity and ω_f denotes foot yaw rate; F_b is the contact force magnitude on body b , $\tau = 0.1$ is the contact threshold, and only contacts outside the allowed hind feet are counted. q_t is the current joint-position vector; q_t^{ref} is the reference joint-position vector; q^{def} is the default joint configuration; and a_t is the current action vector; τ_j^{app} and τ_j^{cmd} denote the applied and commanded torque of joint j , respectively.

TABLE II
TERMINATION TERMS USED DURING TRAINING.

Term	Condition	Threshold
Joint-limit	any joint near its limit	$5^\circ, t > 3$
Hind lift	either hind foot height exceeds the limit	$z_f > 0.06, 3 < t \leq 30$
Abrupt change	any joint position changes too much between steps	$0.5, 3 < t \leq 100$

Notation: z_f is the height of foot f ; and t is the episode step.

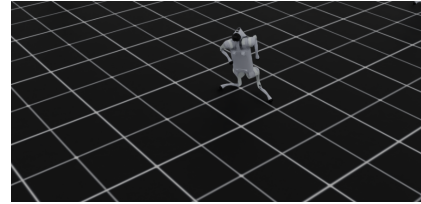


Fig. 2. Snapshot of Stable Bipedal Standing in Simulation

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